School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi
End-semester Examination
Numerical Methods (CS-406)
7^{th} March 2023

Time: 3 hours | Total Marks: 50

PART - A

Answer all questions. (5 × 4 = 20 marks)

Q 1. How many positive roots does the equation $3x3-2x^2-x-5=0$ have?

(a) 2 (b) 3
(c) 1 (d) 0

Q 2. The positive real roots of the equation $x^3-3x+1=0$ lie in the intervals:

(a) (0, 1) & (1,2) (b) (2,1) & (1,3)
(c) (0,2) & (1,3) (d) (0, 1) & (2,3)

Q 3. Using Newton Raphson's method, the the smallest positive root of the equation $f(x) = x^3-5x+1 = 0$ is:

(a) 0.201640 (b) 0.120768
(c) 0.301640 (d) 0.102640

Q 4. The solution of the system of equations

$$x_1 + 10x_2 - x_3 = 3$$

$$2x_1 + 3x_2 + 20x_3 = 7$$

$$10x_1 - x_2 + 2x_3 = 4$$

using the Gauss elimination with partial pivoting.

(a) $x_1 = 0.53752, x_2 = 0.48940., x_3 = 0.36918$
(b) $x_1 = 0.37512, x_2 = 0.28940., x_3 = 0.26908$
(c) $x_1 = 0.17512, x_2 = 0.38194., x_3 = 0.32690$
(d) $x_1 = 0.27432, x_2 = 0.18953., x_3 = 0.13469$

Q.5. Using Lagrange's formula, the quadratic polynomial that takes the following values are:

$$x_0 = 0, y_0 = 0$$

$$x_1 = 1, y_1 = 1$$

$$x_2 = 3, y_2 = 0$$

(a) $6x^2 + 4x - 1$ (b) $4x^2 - 3x + 2$
(c) $\frac{-1}{2}(x^2 - 3x)$ (d) $\frac{1}{2}(3x^2 - x)$

Part-B

Answer all questions. (10 marks each, 10 = 30)

Q.6. The following data gives the velocity of a particle for 20 seconds at an interval of 5 seconds. Find the initial acceleration using the entire data.

Time (sec)	0	5	10	15	20
Velocity (m/sec)	0	3	14	69	228

Q.7. Find an approximation to y(1.6), for the initial value problem

$$y' = x + y^2, y(1) = 1$$

using the Euler method with h = 0.1.

Q.8. Derive the trapezium rule using the Lagrange linear interpolating polynomial. Find the approximate value of

$$I = \int_0^1 f(x)dx$$

$l_0(x) = \frac{x-b}{a-b}$

where

$$f(x) = \frac{1}{1+x}$$

$l_1(x) = \frac{x-a)}{b-a}$

using the trapezium rule with 4 equal subintervals.

$I = \int_a^b f(x)\,dx$